

Histogram Interval Builder

Unit 8: Sports Analytics Lab · Lesson 8-6 · EduWonderLab

UNIT 8 · LESSON 8-6 · TEACHER NOTES

Standard 6.SP.4

FORMAT Bin sort + frequency table

TIME 25–35 min

GROUPING Pairs

TOPIC Display Data: Histograms

STANDARD 6.SP.4

PREP LEVEL Light — print data cards, interval bins, histogram grid

Learning goal *I can sort data into intervals, build a frequency table, and draw a histogram.*

Teacher setup

- Print one set per pair. Students sort data values into equal interval bins.
- They build a frequency table and draw the histogram (bars touch).
- Include a “fix the interval” card with overlapping intervals.

Materials

Data cards, Interval bins, Frequency table, Histogram grid, Pencil.

Differentiation & language support

- Reminder: intervals are equal and do not overlap; bars touch (SPED).
- Sentence stem: “The interval ____ has ____ values.”
- ESOL: tally each value into a bin before counting.

Key vocabulary (English / Español):

Term	Español	What it means
interval	intervalo	an equal range of values
frequency	frecuencia	how many values are in an interval

Quick teacher check: *Check the fix on Card 5 (overlapping at 10).*

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Name: _____ Date: _____ Class: _____

Your goal: *I can sort data into intervals, build a frequency table, and draw a histogram.*

What you need

Data cards, Interval bins, Frequency table, Histogram grid.

How to play

1. Read the data and the intervals on a card.
2. Tally each value into the correct interval.
3. Write the frequency for each interval.
4. Draw the histogram with bars that touch.

Data & Interval Cards

Sort each value into an interval, count the frequency, then build the histogram.

1. Data: 3, 7, 8, 12, 15, 16, 19, 22. Intervals 0–9, 10–19, 20–29.	2. Data: 1, 2, 4, 5, 5, 8, 9. Intervals 0–4, 5–9.
3. Data: 10, 11, 14, 20, 23, 25, 30, 31, 35. Intervals 10–19, 20–29, 30–39.	4. Data: 6, 7, 7, 8, 9, 10, 11, 12, 13. Intervals 6–8, 9–11, 12–14.
5. Fix-it card: Intervals 0–10 and 10–20 overlap at 10. Fix them.	6. Data: 21, 22, 25, 28, 29, 31, 33. Intervals 20–24, 25–29, 30–34.

Frequency table (interval, frequency):

Explain your thinking

Explain why histogram intervals must be equal and cannot overlap.

Sentence frame: *The interval ___ has ___ values because ____.*

★ **Challenge:** *For Card 1, which interval is the tallest bar, and what does that tell you about the data?*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Data: 3, 7, 8, 12, 15, 16, 19, 22. Intervals 0–9, 10–19, 20–29. Answer: 0–9: 3 • 10–19: 4 • 20–29: 1	2. Data: 1, 2, 4, 5, 5, 8, 9. Intervals 0–4, 5–9. Answer: 0–4: 3 • 5–9: 4
3. Data: 10, 11, 14, 20, 23, 25, 30, 31, 35. Intervals 10–19, 20–29, 30–39. Answer: 10–19: 3 • 20–29: 3 • 30–39: 3	4. Data: 6, 7, 7, 8, 9, 10, 11, 12, 13. Intervals 6–8, 9–11, 12–14. Answer: 6–8: 4 • 9–11: 3 • 12–14: 2
5. Fix-it card: Intervals 0–10 and 10–20 overlap at 10. Fix them. Answer: Use 0–9 and 10–19 (no overlap)	6. Data: 21, 22, 25, 28, 29, 31, 33. Intervals 20–24, 25–29, 30–34. Answer: 20–24: 2 • 25–29: 3 • 30–34: 2

Teaching notes & look-fors

- Intervals must be equal width and not overlap; bars touch.
- Card 5: 0–10 and 10–20 both include 10 — fix to 0–9 and 10–19.